GNBF5020

TODAY'S STUDY PLAN

Saturday, 24 October 2026 | Dr. Tony Wong | GNBF5020 Introduction to Molecular Biology and Genetics

What you must MEMORISE today, and what you must STUDY today

[ENGLISH ONLY] ALL EXAM ANSWERS MUST BE IN ENGLISH. Dr. Wong stated this in Lectures 1, 4 and 5 and said he gave zero marks to students who answered in Chinese last year. Every assignment and every exam answer in this course must be written in English.

TIME BUDGET - PICK ONE BEFORE YOU START.

Full day (Blocks 1 to 6, 4 h 30 min of work). The recommended plan; everything in Section 9.

Minimum viable day (90 minutes). Block 1 (the five named questions), the Section 2 numbers, then Block 6 (the 12-question self-test).

In the exam (90 minutes). About 55 minutes for the MCQs, 30 minutes for the short question, 5 minutes to review.

HOW THIS SHEET IS ORGANISED. MEMORISE (Sections 1 to 5) is what must be automatic: the five questions he named, his numbers, the one-liners, the distractor pairs and the answer skeleton. STUDY (Sections 7 and 8) is what must be understood well enough to apply: eight reasoning topics and two scenario algorithms. Section 6 is what NOT to study today. Section 9 is the timetable, Section 10 is the self-test, Section 11 places today inside the 72-hour plan.

WHERE THE MARKS ARE. Multiple choice is for passing; the scenario questions are where the marks are. Plan for 20 MCQ (20 marks) plus two out of three short questions (10 marks) = 30 marks in 90 minutes, and the midterm is 30% of the final grade. Part A is single best option only - last year students lost marks by trying to select multiple options.

Companion document: GNBF5020_今日學習計劃_繁體中文.pdf - Traditional Chinese. All gene, protein, pathway, method and drug names stay in English in both versions.

Video guides

Twelve terms in this book carry the most exam weight, and each one has a short video guide: **English narration, Traditional Chinese subtitles**, about a minute each. The MP4 files live in the project folder **_video/assets/** - the full path is printed next to every link below, so you can still find a file after this PDF has been moved. **_video/index.html** plays all twelve on one page.

Video	Term	Chapter	Video file
V01	**Kaplan–Meier curve**	Ch 10 · Cancer biology and targeted therapy	_video/assets/V01_Kaplan_Meier_curve.mp4
V02	**Contractile ring → binucleated cell**	Ch 8 · Cell biology	_video/assets/V02_Contractile_ring_binucleated_cell.mp4
V03	**NHEJ defect → SCID**	Ch 6 · DNA damage and repair	_video/assets/V03_NHEJ_defect_SCID.mp4
V04	**Mitochondrial pedigree**	Ch 8 · Cell biology	_video/assets/V04_Mitochondrial_pedigree.mp4
V05	**Meiosis I vs Meiosis II (M1 vs M2)**	Ch 8 · Cell biology	_video/assets/V05_Meiosis_I_vs_Meiosis_II_M1_vs_M2.mp4
V06	**BER vs NER**	Ch 6 · DNA damage and repair	_video/assets/V06_BER_vs_NER.mp4
V07	**Hot tumour vs cold tumour**	Ch 9 · Immunology and cancer immunity	_video/assets/V07_Hot_tumour_vs_cold_tumour.mp4
V08	**Driver vs passenger**	Ch 10 · Cancer biology and targeted therapy	_video/assets/V08_Driver_vs_passenger.mp4
V09	**IgG1 vs IgG4**	Ch 9 · Immunology and cancer immunity	_video/assets/V09_IgG1_vs_IgG4.mp4
V10	**Predictive vs prognostic biomarker**	Ch 11 · Assay techniques and omics	_video/assets/V10_Predictive_vs_prognostic_biomarker.mp4
V11	**cis vs trans**	Ch 7 · Control of gene expression	_video/assets/V11_cis_vs_trans.mp4
V12	**de novo vs maintenance methylation**	Ch 4 · Chromatin and epigenetics	_video/assets/V12_de_novo_vs_maintenance_methylation.mp4

Open the video index page ->

0. How to use this sheet

THE SPINE: MEMORISE versus STUDY. MEMORISE means you can produce it without looking - drill it until it is automatic, because the MCQs are pure recall and the clock is short. STUDY means you can use it on a problem you have never seen before - the short questions are won by structure, not by volume. Everything in Sections 1 to 5 is MEMORISE; everything in Sections 7 and 8 is STUDY.

THE TWO SCENARIO FAMILIES. The short questions come from two families only: a pedigree or genetic-counselling report, and a survival curve or dataset. He named both himself. Practise those two, not everything.

THE FORMAT CONFLICT - USE THE LATER STATEMENT. Lecture 4 (10 Oct 2026) said 25 MCQ and choose 1 of 2 short questions; Lecture 5 (17 Oct 2026, later, and written on his own title slide) said 20 MCQ and choose 2 of 3 short questions, 1.5 hours. Plan for Lecture 5. Either way the ratio and the strategy are the same: MCQs are for passing, the scenario questions are where the marks are.

THE ENGLISH RULE, AGAIN. Dr. Wong said this three separate times: "every assignments and exam should be done in English, even this is a Chinese University of Hong Kong" (Lecture 1); "I'm not kidding, because last year there are some student they answered the question in Chinese, and I have no choice that I give the zero mark, because everything should be in English" (Lecture 1); "you have to answer in English, although this is a CUHK, haha, okay" (Lecture 4). This sheet is in English so that you revise in the language you will be graded in. Write every answer in English.

HOW TO USE A SECTION. Read the question first, close the sheet, then answer out loud or on paper. If you cannot produce it, that section is not done. Sections 1 to 5 are done when they are fluent; Sections 7 and 8 are done when you can compare each pair out loud with one clinical consequence.

1. MEMORISE - The five questions he named

1.1 Reading a Kaplan-Meier curve — LECTURE 1

> "And I ask this single question every, every time when I talk about the KM plot, yeah. And only a single student give me a good answer last year, yeah." - Lecture 1

THE QUESTION. From a single Kaplan-Meier plot, deduce early versus late involvement of a gene, and whether the gene separates the two patient groups.

THE ANSWER.

- The curve starts at 1.0 and falls in steps. Flat segments = no events in that interval; a downward step = an event; small tick marks on the curve = censored patients.
- A steep and early drop means the gene is associated with early involvement and a poor prognosis; a late drop means late involvement. Two separated curves = the gene separates the two groups; curves that cross or overlap = the gene does not separate prognosis.
- The lecture's worked example: the same target can be pro-tumour in one cancer and anti-tumour in another. ERBB2 in the Human Protein Atlas is prognostic in renal cancer (favourable), endometrial cancer (unfavourable) and pancreatic cancer (unfavourable), all at $p < 0.001$.

> **[MARK-WINNING LINE]** A steeper and earlier drop indicates early involvement and a worse prognosis; and because the same target can behave in opposite directions in different tumour types - ERBB2 is favourable in renal cancer but unfavourable in endometrial and pancreatic cancer - this finding must not be generalised beyond this tumour type.

1.2 Contractile-ring inhibitor gives a binucleated cell LECTURE 2

> "I remember that I asked this question during the midterm in the first year, yeah. I think only half student can answer it correctly, yeah... Actually quite a surprise to me, yeah. So again this is the type of the question that I would try to include in the midterm, yeah." - Lecture 2

THE QUESTION. You treat dividing cells with an inhibitor of the contractile ring (or knock out its expression). What do you see under the microscope, and what do you see on a flow-cytometry DNA-content plot?

THE ANSWER.

- The contractile ring is actin plus myosin and it drives cytokinesis - the cytoplasmic division.
- Mitosis (chromosome separation) is independent of cytokinesis and proceeds normally. The genetic material replicates and divides, but the cell cannot cleave into two daughter cells.
- The result is one cell with two nuclei - a binucleated cell - with 4N DNA content. Flow cytometry shows a 4N peak instead of separate 2N and 4N peaks. The point of the whole question is that mitosis and cytokinesis are separate processes.

[MARK-WINNING LINE] Blocking the actin-myosin contractile ring stops cytokinesis only, so the cell completes mitosis normally and becomes one binucleated cell with 4N DNA content, because mitosis and cytokinesis are separate processes.

1.3 NHEJ defect gives SCID LECTURE 4

> "Therefore, if I ask you, if the people showing a defective in this non-homologous end joining, what's the outcome? Definitely the outcome is the severe immune deficiency, right? Because he cannot generate a sufficient amount of the antibody pool to recognize the thousand copies of the foreign antigen. Yeah, that's the clinical outcome of this genetic defect." - Lecture 4

THE QUESTION. What is the clinical outcome of a defective non-homologous end joining (NHEJ)?

THE ANSWER.

- NHEJ is co-opted by V(D)J recombination to assemble V + D + J segments. The segment copy numbers he gave are V = 39 copies, D = 27 copies, J = 6 copies.
- That recombination generates antibody (B-cell) and TCR (T-cell) diversity.
- A defect in NHEJ means the repertoire cannot be built, and the clinical outcome is SCID (severe combined immunodeficiency).

[MARK-WINNING LINE] Defective NHEJ breaks V(D)J recombination, so the cell cannot build the antibody and TCR repertoire, and the clinical outcome is severe combined immunodeficiency (SCID).

1.4 Mitochondrial inheritance and the pedigree LECTURE 2

> "So, this is you'll have to a 100% understand, right? Not only the mutation within the mitochondrial DNA can cause the mitochondrial disease. Some mitochondrial disease can actually result from the mutation within the nuclear DNA." - Lecture 2

THE QUESTION. Read a pedigree and decide whether the disease is mitochondrial, and state the recurrence risk for the proband.

THE ANSWER.

- Three hallmarks of maternal (mitochondrial) inheritance: an affected mother passes the condition to all of her children; an affected father passes it to none of his children (sperm mitochondria are destroyed after fertilisation); the condition appears in every generation.
- The twist that catches people: mitochondrial disease is not always mtDNA. Mutations in nuclear genes encoding mitochondrial proteins cause the same phenotype but with a completely different (Mendelian) inheritance pattern, so a mitochondrial-looking pedigree can be nuclear.
- The threshold effect (heteroplasmy): the presence of mutant mtDNA does not guarantee disease. Disease appears only when the proportion of mutant mtDNA exceeds a tissue-specific threshold. High-energy-demand tissues (contracting skeletal muscle) have a low threshold and are affected first; low-demand tissues (hearing) have a high threshold.

> **[MARK-WINNING LINE]** The pedigree shows maternal inheritance - affected mother to all children, affected father to none, and the condition in every generation - but mitochondrial disease can also arise from nuclear gene mutations, so the pedigree cannot be assigned to mtDNA on pattern alone; and the recurrence risk is a probability, not a certainty, because of the heteroplasmy threshold effect.

1.5 M1 versus M2 non-disjunction LECTURE 3

> "And you might need to distinguish the M1 or the M2 failure, because if there's a M1 failure, there's a formation of the germ cell without a losing copy of the chromosome after the M1..." - Lecture 3

THE QUESTION. Distinguish meiosis-I from meiosis-II non-disjunction, and state what gametes come out of one meiosis.

THE ANSWER.

- M1: the homologous chromosomes fail to separate, giving 2 normal + 2 abnormal gametes, all of the abnormal ones the same type. Crossover (recombination) occurs between homologues.
- M2: the sister chromatids fail to separate, giving 1 normal + 3 abnormal gametes (2 of one type, 1 of another). No crossover occurs.
- Related: an inversion changes no genetic material and no gene dosage, yet it can drastically change expression by moving a gene away from its enhancer (or placing an enhancer next to a different promoter).

> **[MARK-WINNING LINE]** In M1 failure the homologous chromosomes fail to separate, giving 2 normal and 2 abnormal gametes with all abnormalities of the same type; in M2 failure the sister chromatids fail to separate, giving 1 normal and 3 abnormal gametes - and because M1 is where crossover occurs and M2 is where it does not, the two are distinguishable on the products alone.

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2. MEMORISE - The numbers

Every value he laboured over with a precise number. These are ready-made MCQ stems; learn the number together with the thing it belongs to.

Table A - DNA, chromatin and structure

Value	What it is for	Lect
G-C = 3 hydrogen bonds; A-T = 2	GC-rich DNA needs a higher Tm to denature - PCR primer design	1, 3
GC content pushes up the melting temperature - more GC means a higher Tm (his slide gives the direction; the notes also carry an "approx 40 deg C drop" and a "60 deg C vs 55 deg C" board example, and the two do not reconcile)	GC content pushes up the melting temperature	3
2.8 A nucleosome crystal structure; below 2.0 A is "very excellent"	X-ray resolution	3
23 pairs of chromosomes; Trisomy 21; Turner 45,X	aneuploidy	3
approx 10 nm nucleosome diameter	"beads on a string"	3
Telomere 3-20 kb TTAGGG; subtelomeric 100-300 kb	chromosome end structure	3
Top 10 % longest telomeres gives worse recurrence-free survival	telomere length in HCC	4

Table B - Cancer biomarkers and statistics

Value	What it is for	Lect
OR 39.6 (CIMP-high vs CIMP-0; 95 % CI 20.6-76.1; p < 0.0001)	CDKN2A methylation	3
OR 12.7 (7.84-20.7; p < 0.0001)	CDKN2A methylation / p16 loss	3
miR-15b + miR-130b: AUC 0.981, sensitivity 98.25 %, specificity 91.53 %, accuracy 94.83 %	circulating miRNA classifier for HCC	1
Serum vimentin: controls 176.0 +/- 61.3; HCC 2 cm or less 271.5 +/- 165.9; HCC more than 2 cm 231.3 +/- 132.3 ng/mL	HCC biomarker	1
Rasonque (daraxonrasib) median OS 13.2 vs 6.7 months	K-Ras G12C inhibitor	1
Anti-TERT antisense oligonucleotide for 3 weeks before implantation gives a tumour-take rate of 2/12, against 9/12 untreated, p = 0.01	HCC tumourigenesis	4
More than 3 HBV integration sites gives significantly poorer survival	HBV integration as a prognostic marker	2
ERBB2 prognostic in renal (favourable), endometrial (unfavourable) and pancreatic (unfavourable), all p < 0.001	Human Protein Atlas	1

Table C - Immunology, cell biology and assay

Value	What it is for	Lect
V = 39 copies, D = 27 copies, J = 6 copies	V(D)J gene segment copies	4
Serum starvation down to 0.5-1 % FBS	G1 arrest and synchronisation	2, 4
15 % of colorectal cancers are MSI-H	MMR deficiency	4
TNF-alpha peaks at 4-6 h, then IL-1 beta, IFN-gamma, IL-2 / IL-15	NF-kappa-B cytokine time course	5
Receptor occupancy 1 % (typical antibody), 2 % (Herceptin in breast), 10 % (Herceptin in gastric)	RO thresholds - the same drug, different dosing, because the required RO is tissue-specific	5
FISH counts gene copies, not antibodies, so it gives HER2 amplification and therefore Herceptin candidacy	companion diagnostic	2

Table D - Clinical trials, pharma and genomics

Value	What it is for	Lect
approx US$200,000 per patient	cost of a targeted therapy	2, 4
Phase 1A dose escalation, Phase 1B expansion, Phase 2/3 efficacy	clinical trial design	2, 4
A repressor encoded on chromosome 1 acting on a target gene on chromosome 13	the cis versus trans example	5
Exon 2 is the diagnostically informative methylation region (Exon 1 is not)	methylation assay design	not stated

3. MEMORISE - The 13 one-liners

One row per Tier-2 concept. Each row is the single discriminating fact you must be able to recall, not a summary of the topic.

Concept	The one fact you must be able to recall	Lect
DNA methylation and epigenetics	CpG methylation suppresses transcription, so promoter hypermethylation of a tumour-suppressor gene is a common cancer driver - it silences the brake. MGMT promoter methylation sensitises glioblastoma to alkylating chemo plus radiotherapy; CIMP predicts a poorer outcome in colorectal cancer.	3
cis versus trans regulation	Cis elements (promoter, enhancer, silencer, insulator) act on the same DNA molecule they sit on; trans factors are diffusible and can act across chromosomes - his example is a repressor on chromosome 1 acting on a target on chromosome 13. Not all enhancers are upstream; some sit at the 3' end of the target gene. Expression is cell-type specific AND time specific.	5
lac operon (two conditions)	Strong transcription needs BOTH lactose present (allolactose binds the Lac repressor, which can then no longer block the operator) AND glucose low (cAMP high, so CAP-cAMP binds the promoter and recruits RNA polymerase). The trp operon is the mirror image: tryptophan binds the Trp repressor as a corepressor and shuts down tryptophan biosynthesis.	5
hot versus cold tumours	MSI-H / MMR-deficient gives frameshift mutations, a high neoantigen burden and a "hot tumour" that responds to anti-PD-1. Cold tumours can be re-programmed hot, for example an HDAC inhibitor inducing PD-L1 - "gene addiction" - and then checkpoint blockade.	5
DNA replication directionality	Nucleotide addition (elongation) is 5' to 3'; proofreading and excision are 3' to 5'; ligase seals 5' phosphate to 3' hydroxyl. The leading strand is continuous with one primer; the lagging strand is discontinuous Okazaki fragments, each needing its own RNA primer. Thousands of origins, each with two bidirectional forks.	4
HR versus NHEJ	HR uses the sister chromatid as a template, is error-free and works only in S/G2; NHEJ needs no template, works in all phases and is error-prone (indels and frameshifts). CRISPR knock-in needs HR; CRISPR knock-out uses NHEJ.	4
Synthetic lethality and PARP inhibitors	BRCA1/2 loss makes the tumour HR deficient, so a PARP inhibitor kills the tumour while normal cells survive on intact HR - a wide therapeutic window, which is the margin between the minimum effective dose and the toxic dose.	4
Nuclear receptor domains, dimerisation, receptor occupancy	NTD (variable, regulatory) / DBD (conserved, binds DNA response elements) / LBD (binds ligand, conformational change, recruits co-activators or co-repressors). Monomers bind DNA poorly; dimerisation dramatically increases affinity. HER2 cannot homodimerise efficiently - it heterodimerises with HER1/HER3/HER4, and Herceptin blocks that.	5
Signal peptide	The N-terminal signal peptide routes the nascent protein into the secretory pathway: rough ER, then Golgi, then plasma membrane or secretion. Removing it blocks expression of secreted and membrane proteins entirely. It is NOT the N-terminal domain (NTD) of a nuclear receptor.	2
Chromatin accessibility	Euchromatin is open and transcriptionally active; heterochromatin is condensed and silent, and it physically blocks transcription factors and enhancers from reaching the promoter. Nucleosomes slide along DNA; two drivers, intrinsic DNA-histone affinity and ATP-dependent chromatin remodelling complexes. Histone-DNA binding is non-covalent, so it is dynamic and reversible.	3
Karyotyping resolution	G-banding detects gross structural change at roughly 0.5-10 Mb; point mutations and single-nucleotide variants need sequencing (Sanger or NGS) and karyotyping misses them. p arm = short, q arm = long, centromere = spindle attachment; it is pronounced "q-two-one-point-one", not "q-twenty-one-point-one".	3
Driver versus passenger	A driver confers a growth advantage; a passenger is hitching a ride, carried along by chromosomal proximity or instability. To find true drivers you must integrate CNV profiles with mRNA expression - concordant copy gain plus transcriptional upregulation (worked example: NPM1 in leukaemia).	3
Flow cytometry fundamentals	FSC = size; SSC = granularity / internal complexity. PI staining quantifies DNA content, giving G1 (2N) / S / G2-M (4N). Serum starvation (FBS down to 0.5-1 %) arrests cells at G1 to synchronise them. FISH uses DNA probes, not antibodies, so it counts gene copies.	2

4. MEMORISE - The 19 distractor pairs

These are the classic "which one is different?" MCQ structures. He keeps returning to each pair, and each pair has exactly one discriminating fact.

Pair	The one discriminating fact
BER versus NER	BER repairs a single damaged base; NER removes bulky multi-nucleotide lesions, for example UV dimers.
MMR versus HR	MMR repairs replication mismatches and small indels; HR is high-fidelity double-strand-break repair.
HR versus NHEJ	HR uses a template and is S/G2 only; NHEJ needs no template, works in all phases and is error-prone.
cis versus trans	cis acts on the same DNA molecule; trans is a diffusible factor encoded elsewhere.
IgG1 versus IgG4	IgG1 = ADCC; IgG4 = no ADCC and a longer half-life.
Autologous versus allogeneic CAR-T	patient-derived versus donor- or iPSC-derived.
In vivo CAR-T	mRNA in lipid nanoparticles, engineered in the body.
Driver versus passenger	causal growth advantage versus bystander.
Euchromatin versus heterochromatin	open and active versus condensed and silent.
DBD versus LBD	DNA-binding domain versus ligand-binding domain.
de novo versus maintenance methylation	establishes a new mark versus copies the mark onto the daughter strand after replication.
Predictive versus prognostic biomarker	predicts response to therapy versus predicts disease outcome.
Dimerisation versus monomer	monomers bind DNA poorly; dimers bind strongly.
5' to 3' versus 3' to 5'	elongation versus proofreading (exonuclease activity).
M1 versus M2 non-disjunction	homologues versus sister chromatids.
mtDNA versus nuclear DNA	maternal inheritance versus Mendelian, with different threshold logic.
ADCC versus CDC	antibody-dependent cellular cytotoxicity versus complement-dependent cytotoxicity.
Signal peptide versus NTD	ER targeting versus nuclear receptor regulatory domain.
Exon 1 versus Exon 2 methylation	Exon 2 is the diagnostically informative region.

5. MEMORISE - The English answer skeleton

A short question is won by structure, not volume. Four moves, every time, in English, in complete sentences.

The four moves

1	State the observation - what the data shows, in one sentence, with the number.
2	Name the mechanism - the pathway or process responsible.
3	Interpret the clinical meaning - what it means for the patient.
4	State the action or the limit - what you would do, or what you cannot conclude.

The generic skeleton - use this for every short question

"From the [plot/table/pedigree], [observation with the specific value]. This is consistent with [mechanism], because [reason]. Clinically this means [interpretation for the patient]. Therefore I would [action]. Note that we cannot conclude [limitation], because [reason]."

The Kaplan-Meier plot template

"The curve for [group A] drops from 1.0 to [x] within [n] months, whereas [group B] remains near [y] until [m] months. The [log-rank] p-value is [p], so the two groups are separated. A steeper and earlier drop indicates [early involvement / worse prognosis]. This is consistent with [mechanism]. The same target can behave in opposite directions in different tumour types - [ERBB2 example] - so this finding should not be generalised beyond [this tumour type]. The practical implication is [action]."

The pedigree template

"The pedigree shows [transmission pattern], which is consistent with [mitochondrial / autosomal dominant / X-linked] inheritance. The three features supporting this are [feature 1], [feature 2], [feature 3]. Importantly, [mitochondrial disease can also arise from nuclear gene mutations], so the pedigree cannot be assigned to mtDNA on pattern alone; confirmation would require [sequencing both the mitochondrial genome and relevant nuclear genes]. The recurrence risk for the proband is [x]%. This is a probability, not a certainty, because of [incomplete penetrance / threshold effect / variable expression]."

Marker-pleasing habits

- State the number. A number in the answer is worth more than a paragraph without one.
- Name both the mechanism AND the gene or protein. "PARP inhibitor" alone is half an answer; "a PARP inhibitor, because the tumour is BRCA1/2-deficient and therefore HR-deficient" is the answer.
- State a limitation. Every one of his templates ends by saying what you cannot conclude.
- Write complete sentences. No bullet fragments - he is marking your reasoning, in English.
- Never, ever answer in Chinese. Zero marks.

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6. DO NOT STUDY TODAY

SKIP LIST - he put these out of scope, so do not revise them

Not examinable	His words	Lect
~~Chemical structures and metabolic pathway details~~	*"Please don't memorize all these chemical structure... They would not be included in my exam."*	1
~~The amino acid R group table~~	*"don't memorize this... definitely not include in the exam"*	1
~~FISH protocol parameters (denaturation, temperature)~~	*"Definitely not include in the exam. I will not ask you about what's the denaturations, what's the temperature"*	2
~~Specific cyclin CDK pairings~~	*"you don't need to memorize all this cyclin or cyclin-dependent kinase, CDK"*	2
~~Detailed signalling cascades on the slide~~	*"I don't memorize all this thing, I just highlight here in case some student are really interested in molecular biology"*	2
~~ADC structures and dual payload ADCs~~	*"all this would not be included in the midterm. I'm just want you to know more"*	4
~~Pharma target nomination and the druggability workflow~~	*"it's not included in exam, but I think it's really more interesting"*	4
~~DNA glycosylase and abasic (AP) site chemistry~~	*"I can ensure that they will not be included in the coming midterm"*	4

~~CNV molecular mechanisms in depth~~	*"I'm just give you a taste... don't be scared with all these mechanism"*	3
~~Electron microscopy chromatin images~~	*"don't memorize all these figures, yeah. It's not necessary"*	3
~~The exact cytokine sequence~~	*"don't worry about all these cytokines. I'm just want to show you that different responses can be observed in different times."*	5

7. STUDY - The eight reasoning topics

7.1 HR versus NHEJ, and the SCID chain — LECTURE 4

THE EXAM QUESTION IT ANSWERS. "What is the consequence of inactivating X?" - defective NHEJ gives SCID. And "Which editing strategy would you choose for a knock-in?"

WHAT UNDERSTANDING LOOKS LIKE. You can say out loud that HR needs a sister-chromatid template, is error-free and is confined to S/G2, while NHEJ needs no template, works in all phases and is error-prone - and then close the loop: NHEJ is co-opted by V(D)J recombination, so a defect destroys the antibody and TCR repertoire and the clinical outcome is SCID. The clinical consequence: a knock-in needs HR and so must be done in S/G2, a knock-out can use NHEJ, and PARP inhibitors work only where HR is already lost.

7.2 lac and trp operon logic — LECTURE 5

THE EXAM QUESTION IT ANSWERS. "Reading the lac operon logic correctly" - which condition is required, and what happens when the second one fails.

WHAT UNDERSTANDING LOOKS LIKE. You state that BOTH conditions are required for strong transcription - lactose present, so allolactose inactivates the Lac repressor, AND glucose low, so cAMP is high and CAP-cAMP recruits RNA polymerase - and that trp is the mirror image, with tryptophan acting as a corepressor. The clinical consequence: IPTG is the artificial, non-metabolised inducer used in E. coli recombinant systems, and CHO cells, which require codon optimisation, are used when glycosylation is required.

7.3 Hot versus cold tumour, and checkpoint blockade - Lectures 4 and 5

THE EXAM QUESTION IT ANSWERS. "A patient with X is most likely to respond to which drug?" - MSI-H to anti-PD-1.

WHAT UNDERSTANDING LOOKS LIKE. You can chain it out loud: MSI-H / MMR-deficient gives frameshift mutations, which gives a high neoantigen burden, which gives a hot tumour, which responds to anti-PD-1 - and you know that the dMMR versus pMMR Kaplan-Meier curve is the figure he calls one of the most important figures in modern medicine. The clinical consequence: cold tumours can be re-programmed hot, for example by an HDAC inhibitor inducing PD-L1, so "not a hot tumour" is a target, not a dead end.

7.4 Synthetic lethality and PARP inhibitors — LECTURE 4

THE EXAM QUESTION IT ANSWERS. "A BRCA-mutant patient: which drug, and why is it selective for tumour cells?"

WHAT UNDERSTANDING LOOKS LIKE. You can say BRCA1/2 loss makes the tumour HR deficient, so a PARP inhibitor kills the tumour while normal cells, with intact HR, survive - and you can explain that this widens the therapeutic window, the margin between the minimum effective dose and the toxic dose. The clinical consequence: it is a tumour-selective drug rather than a general cytotoxic, which is why the window is wide enough to use.

7.5 Chromatin accessibility and epigenetics

LECTURE 3

THE EXAM QUESTION IT ANSWERS. "Why is this promoter silent?" and "Is this driver or passenger?"

WHAT UNDERSTANDING LOOKS LIKE. You distinguish open euchromatin from condensed heterochromatin that physically blocks transcription factors and enhancers from reaching the promoter, and you explain that nucleosome sliding is driven by intrinsic DNA-histone affinity plus ATP-dependent remodelling complexes, with non-covalent histone-DNA binding making it dynamic and reversible. The clinical consequence: promoter hypermethylation of a tumour suppressor is a common cancer driver, so a methylated MGMT promoter sensitises glioblastoma to alkylating chemotherapy plus radiotherapy.

7.6 cis versus trans regulation

LECTURE 5

THE EXAM QUESTION IT ANSWERS. "Does this regulatory sequence act on the gene next to it, or somewhere else?"

WHAT UNDERSTANDING LOOKS LIKE. You state that cis elements act on the same DNA molecule, that trans factors are diffusible and can act across chromosomes (the repressor on chromosome 1 acting on a target on chromosome 13), that not all enhancers are upstream because some sit at the 3' end of the target gene, and that expression is cell-type specific AND time specific. The clinical consequence: an inversion can change expression without changing dosage, by moving a gene away from its enhancer.

7.7 Signal peptide and the secretory pathway

LECTURE 2

THE EXAM QUESTION IT ANSWERS. "Why is my recombinant protein not expressed?" and "what is this sequence at the N-terminus?"

WHAT UNDERSTANDING LOOKS LIKE. You say that the N-terminal signal peptide routes the nascent protein into the secretory pathway - rough ER, Golgi, then plasma membrane or secretion - and that removing it blocks expression of secreted and membrane proteins entirely; his postdoc lost four months of work that way. The clinical consequence: it is not the same thing as the N-terminal domain (NTD) of a nuclear receptor, and if a therapeutic protein is not being expressed you check this before you blame the host cell.

7.8 Predictive versus prognostic biomarker - Lectures 1 and 4

THE EXAM QUESTION IT ANSWERS. "Is this marker predicting response to a therapy, or predicting the disease outcome?"

WHAT UNDERSTANDING LOOKS LIKE. You separate a predictive biomarker, which predicts response to therapy, from a prognostic biomarker, which predicts disease outcome regardless of therapy, and you apply the distinction: ERBB2 is prognostic in renal (favourable), endometrial (unfavourable) and pancreatic (unfavourable) cancer, all $p < 0.001$, while MGMT promoter methylation is predictive of alkylating-chemo sensitivity. The clinical consequence: the same marker can be prognostic in one tumour type and predictive in another, so you must always name the tumour type.

8. STUDY - Two scenario algorithms

8.1 Algorithm A - reading a Kaplan-Meier curve (6 steps)

1	Read the axes and the units. x-axis = months; y-axis = probability of survival or progression-free survival. The curve starts at 1.0 and falls in steps.
2	Locate the median survival, and locate the point at which the two curves separate.
3	Identify which group drops first, and how steeply it drops.
4	Account for censoring. The small tick marks on the curve are censored patients, not events, and you must read the number at risk before comparing the curves.
5	Decide early versus late involvement. A steep and early drop means the gene is associated with early involvement and a poor prognosis; a late drop means late involvement; curves that cross or overlap do not separate prognosis at all.
6	State the clinical action AND the limit. Say what you would do for the patient, then say what you cannot conclude, using the ERBB2 example.

[WORKED MICRO-EXAMPLE] Two groups of 40 patients; x = months 0 to 60; y = survival probability. Group A falls from 1.0 to 0.55 within the first 8 months and then flattens; group B stays near 0.9 until month 30 and then falls to 0.45 by month 50. Group A carries 6 tick marks and group B carries 4.

[WHAT YOU WRITE] Group A is the early, steep drop, so the gene separates the two groups and is associated with early involvement and a worse prognosis; the tick marks are censored patients and not events, so the number at risk has to be read before the two groups are compared; and because the same target can behave in opposite directions in different tumour types - ERBB2 - the finding cannot be generalised beyond this tumour type.

THE SENTENCE FRAME YOU WRITE - fill in the Section 5 KM-plot template.

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"The curve for [group A] drops from 1.0 to [x] within [n] months, whereas [group B] remains near [y]
until [m] months. The [log-rank] p-value is [p], so the two groups are separated. A steeper and
earlier drop indicates [early involvement / worse prognosis]. This is consistent with [mechanism].
The same target can behave in opposite directions in different tumour types - [ERBB2 example] - so
this finding should not be generalised beyond [this tumour type]. The practical implication is
[action]."
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8.2 Algorithm B - reading a pedigree (6 steps)

1	Read the transmission pattern across the three generations, and write the pattern down in words before you interpret it.
2	Test the three maternal hallmarks: an affected mother passes the condition to all of her children; an affected father passes it to none; the condition appears in every generation.
3	Check for male-to-male transmission. If an affected father has an affected son, X-linked is ruled out.
4	Consider nuclear versus mitochondrial. Nuclear-gene mutations give the same phenotype with a Mendelian pattern, so the pattern alone cannot assign the pedigree to mtDNA.
5	Apply penetrance and threshold. Heteroplasmy means disease appears only above a tissue-specific mutant fraction; high-energy tissues such as contracting skeletal muscle have a low threshold and go first, low-demand tissues such as hearing have a high threshold.
6	Give the recurrence risk as a probability, not a certainty, and say why.

[WORKED MICRO-EXAMPLE] Generation I: an unaffected man, married in. Generation II: an affected woman married to an unaffected man, with three children - two affected sons and one unaffected daughter. Generation III: each affected son has one unaffected child.

[WHAT YOU WRITE] The pedigree shows a mitochondrial pattern: an affected mother transmits to 2 of her 3 children, an affected father transmits to none, and the trait appears in two consecutive generations. But it cannot be assigned to mtDNA on pattern alone, because mutations in nuclear genes encoding mitochondrial proteins give the same phenotype with a Mendelian pattern, so confirmation requires sequencing both the mitochondrial genome and the relevant nuclear genes. There is no male-to-male transmission, so X-linked is ruled out. The recurrence risk for the proband is a probability and not a certainty, because of incomplete penetrance, variable expression and the heteroplasmy threshold effect.

THE SENTENCE FRAME YOU WRITE - fill in the Section 5 pedigree template.

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"The pedigree shows [transmission pattern], which is consistent with [mitochondrial / autosomal
dominant / X-linked] inheritance. The three features supporting this are [feature 1], [feature 2],
[feature 3]. Importantly, [mitochondrial disease can also arise from nuclear gene mutations], so the
pedigree cannot be assigned to mtDNA on pattern alone; confirmation would require [sequencing both
the mitochondrial genome and relevant nuclear genes]. The recurrence risk for the proband is [x]%.
This is a probability, not a certainty, because of [incomplete penetrance / threshold effect /
variable expression]."
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9. TODAY'S TIMETABLE

Relative blocks, not clock times - start whenever you start. Every block ends with an output you must be holding in your hand.

Block	Length	Type	Content	Output you must have
1	45 min	MEMORISE	Section 1 - the five named questions, closed book first	you can recite all five answers, each with its mark-winning line
2	45 min	MEMORISE	Section 2 numbers plus Section 4 pairs, as flashcards	every number and every discriminator retrievable
3	45 min	STUDY	Section 8 - write both algorithms out by hand	two numbered algorithms on paper, from memory
4	45 min	STUDY	Section 7 topics 1 to 4	you can compare each pair out loud
5	45 min	STUDY	Section 7 topics 5 to 8	you can compare each pair out loud
6	45 min	STUDY	Section 10 - the 12-question self-test, closed book, IN ENGLISH	self-test answered and marked
7 (optional)	60 min	STUDY	re-do 4 highest-yield scenarios from the case bank, timed, written in English	4 written English answers
8 (optional)	45 min	MEMORISE	Section 6 - write the one-page "will NOT be asked" sheet; then read your Block 7 answers aloud for grammar	1 page plus a grammar pass

MINIMUM VIABLE DAY - IF YOU ONLY HAVE 90 MINUTES. Block 1 (the five named questions), then the Section 2 numbers, then Block 6 (the 12-question self-test). That is the floor: the five questions he named, his numbers, and a closed-book English rehearsal.

IN THE EXAM - 90 MINUTES. About 55 minutes for the MCQs, which are the passing floor and pure recall; 30 minutes for the short question, with the four-move template; 5 minutes to review.

10. SELF-TEST - 12 questions

[ENGLISH ONLY] ALL ANSWERS IN ENGLISH. Dr. Wong said this in Lectures 1, 4 and 5 and said he gave zero marks to students who answered in Chinese last year. Closed book, full sentences, every answer.

ANSWER IN ENGLISH. CLOSED BOOK. Write full sentences, state the number, and name both mechanism and gene or protein. No Chinese.

1	KM curve: what does a flat segment mean? What do the tick marks mean? Of two separated curves, which has the worse prognosis, and why?
2	Contractile-ring inhibitor: what is the DNA content of the resulting cell, and what does it prove about mitosis versus cytokinesis?
3	NHEJ defect: what is the clinical outcome, and which step in the chain fails?
4	Mitochondrial pedigree: state the three hallmarks, and say what a nuclear-gene mutation changes about the answer.
5	M1 versus M2: how many normal gametes come from one meiosis with M2 failure, and what fails to separate in each case?
6	Which of these will NOT be examined: (a) chemical structures of metabolic pathways (b) the concept of base-excision repair (c) FISH denaturation temperature - and why?
7	HR versus NHEJ: template? cell-cycle phase? error-free?
8	lac operon: state the two conditions required for strong transcription.
9	MSI-H / MMR-deficient tumour: which drug class, and why does it respond?
10	BRCA-mutant patient: which drug, and why is it selective for tumour cells?
11	Numbers: give three of these from memory - OR 39.6, the V / D / J copy numbers, and Herceptin receptor occupancy in breast versus gastric.
12	Write the 4-move skeleton you will use for every short answer in English.

Answer key

No.	Answer
1	Flat = no events in that interval; a downward step = an event; tick marks = censored patients, not events. The steeper and earlier drop has the worse prognosis, because early involvement is associated with poor outcome; but the same target can behave in opposite directions in different tumour types (ERBB2), so it must not be generalised beyond this tumour type.
2	4N DNA content in one cell with two nuclei. It proves that mitosis and cytokinesis are separate processes: blocking the actin-myosin contractile ring stops cytokinesis only, while chromosome separation proceeds normally. Flow cytometry shows a 4N peak instead of separate 2N and 4N peaks.
3	Severe combined immunodeficiency (SCID). The failing step is V(D)J recombination, which is co-opted from NHEJ: without NHEJ the cell cannot assemble V + D + J segments and so cannot build the antibody and TCR repertoire. Segment copies: V = 39, D = 27, J = 6.
4	Affected mother to all children; affected father to none; the condition appears in every generation. A nuclear-gene mutation gives the same phenotype with a Mendelian pattern, so the pedigree cannot be assigned to mtDNA on pattern alone, and confirmation requires sequencing both the mitochondrial genome and the relevant nuclear genes. Risk is a probability because of heteroplasmy and the tissue threshold.
5	M2 gives 1 normal and 3 abnormal gametes (2 of one type, 1 of another); M1 gives 2 normal and 2 abnormal, all abnormal the same type. M1: homologous chromosomes fail to separate. M2: sister chromatids fail to separate, and no crossover occurs in M2.
6	(a) and (c) will not be examined - he explicitly excluded chemical structures of metabolic pathways and FISH protocol parameters. (b) WILL be examined in concept: "don't memorize the glycosylase chemistry" does not mean "don't know what BER repairs", and BER is in the distractor pairs.
7	HR: uses the sister chromatid as a template, S/G2 only, error-free. NHEJ: no template, all phases, error-prone (indels and frameshifts).
8	Lactose present, so allolactose binds the Lac repressor and it can no longer block the operator; AND glucose low, so cAMP is high and CAP-cAMP binds the promoter and recruits RNA polymerase.
9	Anti-PD-1 (checkpoint blockade). MSI-H / MMR deficiency gives frameshift mutations, a high neoantigen burden and a "hot tumour", so the tumour is visible to the immune system.
10	A PARP inhibitor. BRCA1/2 loss makes the tumour HR deficient, so PARP inhibition kills it, while normal cells with intact HR survive - synthetic lethality widens the therapeutic window between effective and toxic doses.
11	OR 39.6 (CIMP-high vs CIMP-0, 95 % CI 20.6-76.1, $p < 0.0001$). V = 39, D = 27, J = 6 copies. Receptor occupancy 1 % typical antibody, 2 % for Herceptin in breast cancer, 10 % for Herceptin in gastric cancer - the same drug, different dosing, because the required RO is tissue-specific.

No.	Answer
12	1. State the observation, with the number. 2. Name the mechanism. 3. Interpret the clinical meaning. 4. State the action or the limit. Written as: "From the [plot/table/pedigree], [observation with the value]. This is consistent with [mechanism], because [reason]. Clinically this means [interpretation]. Therefore I would [action]. Note that we cannot conclude [limitation], because [reason]."

11. Where today slots into the 72 hours

Today's blocks are drawn from the 72-hour plan. Say plainly what is finished today and what must wait for tomorrow.

72-hour block	Time	Content	Done today?	Where it is in this sheet
A	3 h	Tier 1 in full - KM curves, contractile ring, NHEJ to SCID, mitochondrial pedigree, M1/M2	DONE today	Block 1, Section 1
B	2 h	Tier 2 - each concept, the discriminating fact, one clinical consequence	DONE today	Blocks 4 and 5, Sections 3 and 7
C	1.5 h	The numbers and the distractor pairs, drilled as flashcards	DONE today	Block 2, Sections 2 and 4
D	2 h	Re-do the highest-yield scenarios, in English, in writing, timed	PARTLY - only with optional Block 7	Section 8
E	1 h	The exclusion list - write a one-page "will NOT be asked" sheet	DONE today	Block 8, Section 6
F	1 h	Read your own answers aloud for grammar and structure	DONE today	Block 8, Section 5 and Section 10

THE PLAIN VERSION. Today completes A, B, C and E outright. D and F are completed only if you do the optional Blocks 7 and 8; if you only have the 90-minute minimum viable day, then A and C plus the self-test are what you get, and D and F must wait for tomorrow.

WHAT TOMORROW OWES YOU. The scenario questions are worth 10 of the 30 marks and they are written, not recalled - so tomorrow belongs to Block D, and to the grammar pass in Block F. Today you have built the recall; tomorrow you build the writing.

Figure E3 - The day map, one page overleaf: the eight blocks, the MEMORISE / STUDY / OPTIONAL split, the 90-minute minimum viable day and the in-exam budget.

Figure E3 - The day map



Figure E3. The eight relative blocks of today's plan: the MEMORISE blocks (teal), the STUDY blocks (navy) and the two optional extension blocks (amber, dashed). The 90-minute minimum viable day is the floor if time is short; the MCQ is the passing floor and the scenario questions carry the marks. Figure authored at 6.5 in print width and placed 1:1 on this page.